**Q1: 24 Feb (Shift 1) - Single Correct**

The value of $-{}^{15}\mathrm{C}_1 + 2.{}^{15}\mathrm{C}_2 - 3.{}^{15}\mathrm{C}_3 + \ldots . - 15.{}^{15}\mathrm{C}_{15} + {}^{14}\mathrm{C}_1 + {}^{14}\mathrm{C}_3 + {}^{14}\mathrm{C}_5 + \ldots . + {}^{14}\mathrm{C}_{11}$ is:

(1) $2^{14}$

(2) $2^{13} - 13$

(3) $2^{16} - 1$

(4) $2^{13} - 14$

**Q2: 24 Feb (Shift 2) - Single Correct**

If $n \geq 2$ is a positive integer, then the sum of the series

${}^{\mathrm{n+1}}\mathrm{C}_2 + 2\left({}^2\mathrm{C}_2 + {}^3\mathrm{C}_2 + {}^4\mathrm{C}_2 + \ldots + {}^{\mathrm{n}}\mathrm{C}_2\right)$ is :

(1) $\frac{\mathrm{n(n+1)^2(n+2)}}{12}$

(2) $\frac{\mathrm{n(n-1)(2n+1)}}{6}$

(3) $\frac{\mathrm{n(n+1)(2n+1)}}{6}$

(4) $\frac{\mathrm{n(2n+1)(3n+1)}}{6}$

**Q3: 24 Feb (Shift 2) - Numerical**

For integers $n$ and $r$, let $\binom{n}{r} = \begin{cases} {}^nC_r, & \text{if } n \geq r \geq 0 \\ 0, & \text{otherwise} \end{cases}$

The maximum value of k for which the sum

$\sum_{i=0}^{k} \binom{10}{i}\binom{15}{k-i} + \sum_{i=0}^{k+1} \binom{12}{i}\binom{13}{k+1-i}$ exists, is equal to

Note: NTA has dropped this question in the final official answer key.

**Q4: 25 Feb (Shift 2) - Numerical**

If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + \mathrm{x})^{2022}$ is divided by 8 is

**Q5: 26 Feb (Shift 1) - Single Correct**

The maximum value of the term independent of 't' in the expansion of $\left(tx^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{10}$ where $x \in (0,1)$ is:

(1) $\frac{10!}{\sqrt{3}(5!)^2}$

(2) $\frac{2.10!}{3(5!)^2}$

(3) $\frac{10!}{3(5!)^2}$

(4) $\frac{2.10!}{3\sqrt{3}(5!)^2}$

**Q6: 26 Feb (Shift 1) - Numerical**

Let $m, n \in N$ and gcd $(2, n) = 1$. If $30\binom{30}{0}+29\binom{30}{1}+\ldots.+2\binom{30}{28}+1\binom{30}{29}=\mathrm{n}\cdot 2^{\mathrm{m}}$, then $\mathrm{n}+\mathrm{m}$ is equal to

# Answer Key

**Q1** (4) **Q2** (3) **Q3** (12) **Q4** (1)

**Q5** (4) **Q6** (45)

**Q1 (4)**

$S_1 = -{}^{15}C_1 + 2 \cdot {}^{15}C_2 - \ldots\ldots\ldots\ldots -15{}^{15}C_{15}$

$= \sum_{r=1}^{15}(-1)^r \cdot r{}^{15}C_r = 15\sum_{r=1}^{15}(-1)^{r\,14}C_{r-1}$

$= 15\left(-{}^{14}C_0 + {}^{14}C_1 - \ldots\ldots - {}^{14}C_{14}\right) = 15(0) = 0$

$S_2 = {}^{14}C_1 + {}^{14}C_3 + \ldots\ldots + {}^{14}C_{11}$

$= \left({}^{14}C_1 + {}^{14}C_3 + \ldots\ldots + {}^{14}C_{11} + {}^{14}C_{13}\right) - {}^{14}C_{13}$

$= 2^{13} - 14$

$= S_1 + S_2 = 2^{13} - 14$

**Q2 (3)**

${}^2C_2 = {}^3C_3$

$S = {}^3C_3 + {}^3C_2 + \ldots\ldots + {}^nC_2 = {}^{n+1}C_3$

$\because {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

$\therefore {}^{n+1}C_2 + {}^{n+1}C_3 + {}^{n+1}C_3 = {}^{n+2}C_3 + {}^{n+1}C_3$

$= \frac{(n+1)!}{3!(n-1)!} + \frac{(n+1)!}{3!(n-2)!}$

$= \frac{(n+2)(n+1)n}{6} + \frac{(n+1)(n)(n-1)}{6} = \frac{n(n+1)(2n+1)}{6}$

**Q3 (12)**

$(1+x)^{10} = {}^{10}C_0 + {}^{10}C_1x + {}^{10}C_2x^2 + \ldots\ldots + {}^{10}C_{10}x^{10}$

$(1+x)^{15} = {}^{15}C_0 + {}^{15}C_1x + \ldots.\ {}^{15}C_{k-1}x^{k-1} + {}^{15}C_kx^k + {}^{15}C_{k+1}x^{k+1} + \ldots..\ {}^{15}C_{15}x^{15}$

$\sum_{i=0}^{k}(10C_i)(15C_{k-1}) = {}^{10}C_0 \cdot {}^{15}C_k + {}^{10}C_1 \cdot {}^{15}C_{k-1} + \ldots.. + {}^{10}C_k \cdot {}^{15}C_0$

Coefficient of $x_k$ in $(1+x)^{25}$

$= {}^{25}C_k$

$\sum_{i=0}^{k+1}\left({}^{12}C_i\right)\left({}^{13}C_{k+1-i}\right) = {}^{12}C_0 \cdot {}^{13}C_{k+1} + {}^{12}C_1 \cdot {}^{13}C_k + \ldots\ldots + {}^{12}C_{k+1} \cdot {}^{13}C_0$

Coefficient of $x^{k+1}$ in $(1+x)^{25}$

$= {}^{25}C_{k+1}$

${}^{25}C_k + {}^{25}C_{k+1} = {}^{26}C_{k+1}$

For maximum value

$k+1=13$

K $=12$

**Q4 (1)**

Let $x=4k+3$
$(2020+x)^{2022}$

$=(2020+4k+3)^{2022}$

$=(4(505)+4k+3)^{2022}$
$=(4P+3)^{2022}$

$=(4P+4-1)^{2022}$

$=(4A-1)^{2022}$
${}^{2022}\mathrm{C}_0(4\ \mathrm{A})^0(-1)^{2022}+{}^{2022}\mathrm{C}_1(4\ \mathrm{A})^1(-1)^{2021}+\ldots\ldots$

$1+8\lambda$

Reminder is 1

**Q5 (4)**

$T_{r+1}={}^{10}C_r\left(tx^{1/5}\right)^{10-r}\left[\frac{(1-x)^{1/10}}{t}\right]^r$
$={}^{10}C_r t^{(10-2r)}\times x^{\frac{10-r}{5}}\times(1-x)^{\frac{r}{10}}$

$\Rightarrow 10-2r=0\Rightarrow r=5$

$T_6={}^{10}C_5 x\sqrt{1-x}$
$\frac{dT_6}{dx}={}^{10}C_5\left[\sqrt{1-x}-\frac{x}{2\sqrt{1-x}}\right]=0$

$=1-x=x/2\Rightarrow 3x=2$

$\Rightarrow x=2/3$

$T_6|_{\max}=\frac{10!}{5!5!}\times\frac{2}{3\sqrt{3}}$

**Q6 (45)**

Let $S=\sum_{r=0}^{30}(30-r)^{30}C_r$
$=30\sum_{r=0}^{30}{}^{30}C_r-\sum_{r=0}^{30}r^{30}C_r$

$= 20 \times 2^{30} - \sum_{r=1}^{30} r \cdot \frac{30}{4} \cdot {}^{29}\mathrm{C}_{r-1}$

$= 30 \times 2^{30} - 30 \cdot 2^{29}$

$= (30 \times 2 - 30) \cdot 2^{29} = 30 \cdot 2^{29} \Rightarrow 15 \cdot 2^{30}$

$= n = 15, \quad m = 30$

$n + m = 45$